COMMUTATIVE DIAGRAMS IN THE LOCALIZED CATEGORY COME FROM DIAGRAMS IN C AFTER ENLARGING ALONG S

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Editorial note. Counted span is the author's own proof environment, source lines 4703-4753, following the statement at 4680-4701.

Editorial note. The footnote at 4723-4728 gives an explicit alternative path for matching b = t^{-1}h; it lies inside the counted span.

Allowed standard prerequisites (editorial).

- Calculus of fractions: morphisms and composition. Definition of morphisms and composition in S^{-1}C in this chapter (left calculus of fractions section), used at proof lines 4704-4741. Writes a and b as fractions s^{-1}g and t^{-1}h and computes equivalence classes of composites as pairs.
- LMS2 axiom. Left multiplicative system axiom LMS2, used at 4706-4707 and 4726-4727. Produces the commutative square completing f" over t, and the alternative replacement path in the footnote.
- Equality criterion for fractions. Lemma lemma-equality-morphisms-left-localization of this chapter, invoked at 4747-4750. Turns equality of equivalence classes of pairs into a single post-composition d with d t in S making the arrows equal.
- Replacement remark. Remark remark-left-localization-morphisms-colimit of this chapter, invoked at 4720-4722. Licenses the repeated replacement of (t, f") by postcomposition with d having d t in S.

Source reference. Original lemma 04VF: sources/stacks-categories/pinned-original-source.tex, source lines 4521–4521. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 05Q0: sources/stacks-categories/pinned-original-source.tex, source lines 4580–4580. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:categories.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{C}$ be a category. Let S be a left multiplicative system. If $f : X \to Y$, $f' : X' \to Y'$ are two morphisms of $\mathcal{C}$ and if*

$$\begin{array}{ccc} Q(X) & \xrightarrow{a} & Q(X') \\ {\scriptstyle Q(f)}\downarrow & & \downarrow{\scriptstyle Q(f')} \\ Q(Y) & \xrightarrow{b} & Q(Y') \end{array}$$

is a commutative diagram in $S^{-1}\mathcal{C}$, then there exist a morphism $f'' : X'' \to Y''$ in $\mathcal{C}$ and a commutative diagram

$$\begin{array}{ccccc} X & \xrightarrow{g} & X'' & \xleftarrow{s} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f''} & & \downarrow{\scriptstyle f'} \\ Y & \xrightarrow{h} & Y'' & \xleftarrow{t} & Y' \end{array}$$

in $\mathcal{C}$ with $s, t \in S$ and $a = s^{-1}g$, $b = t^{-1}h$.

Proof. We choose maps and objects in the following way: First write $a = s^{-1}g$ for some $s : X' \to X''$ in S and $g : X \to X''$. By LMS2 we can find $t : Y' \to Y''$ in S and $f'' : X'' \to Y''$ such that

$$\begin{array}{ccccc} X & \xrightarrow{g} & X'' & \xleftarrow{s} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f''} & & \downarrow{\scriptstyle f'} \\ Y & & Y'' & \xleftarrow{t} & Y' \end{array}$$

commutes. Now in this diagram we are going to repeatedly change our choice of

$$X'' \xrightarrow{f''} Y'' \xleftarrow{t} Y'$$

by postcomposing both t and f'' by a morphism $d : Y'' \to Y'''$ with the property that $d \circ t \in S$. According to Remark 05Q0 we may after such a replacement assume that there exists a morphism $h : Y \to Y''$ such that $b = t^{-1}h$ holds[1]. At this point we have everything as in the lemma except that we don't know that the left square of the diagram commutes. But the definition of composition in $S^{-1}\mathcal{C}$ shows that $b \circ Q(f)$ is the equivalence class of the pair $(h \circ f : X \to Y'', t : Y' \to Y'')$ (since b is the equivalence class of the pair $(h : Y \to Y'', t : Y' \to Y'')$, while $Q(f)$ is the equivalence class of the pair $(f : X \to Y, \mathrm{id} : Y \to Y)$), while $Q(f') \circ a$ is the equivalence class of the pair $(f'' \circ g : X \to Y'', t : Y' \to Y'')$ (since a is the equivalence class of the pair $(g : X \to X'', s : X' \to X'')$, while $Q(f')$ is the equivalence class of the pair $(f' : X' \to Y', \mathrm{id} : Y' \to Y')$). Since we know that $b \circ Q(f) = Q(f') \circ a$, we thus conclude that the equivalence classes of the pairs $(h \circ f : X \to Y'', t : Y' \to Y'')$ and $(f'' \circ g : X \to Y'', t : Y' \to Y'')$ are equal. Hence using Lemma 04VF we can find a morphism $d : Y'' \to Y'''$ such that $d \circ t \in S$ and $d \circ h \circ f = d \circ f'' \circ g$. Hence we make one more replacement of the kind described above and we win. □

[1]Here is a more down-to-earth way to see this: Write $b = q^{-1}i$ for some $q : Y' \to Z$ in S and some $i : Y \to Z$. By LMS2 we can find $r : Y'' \to Y'''$ in S and $j : Z \to Y'''$ such that $j \circ q = r \circ t$. Now, set $d = r$ and $h = j \circ i$.